# Euler-Zeno in Prime Relatives (EZPR)

An Infinite-Address Relative-Coordinate Model
Virgil Lee Gattenby

## Abstract

I draw a line. I identify its ends and compare a point's position from each side. When both readings use the same span and match, the point can only be halfway between them.

Now I give that construction a family of numbers. I remove zero, one and every composite from the candidate list. What remains is exactly the primes. Each prime retains its identity while its admitted matching event occupies the midpoint of its parent span. Sharing a location does not make them the same number.

I call this Euler-Zeno in Prime Relatives (EZPR). Infinite addressing describes the entire family through one rule and one argument, without plotting every member separately. The proofs establish midpoint placement under matching, preservation through changes of scale and successive generations, and preservation of inherited distinctions through exact recovery. Lean verifies the arithmetic selection and matching-based placement results.

The principal result is proved local closure of EZPR: every admitted prime event has its unique parent-relative coordinate ½, and no admitted prime event lies off that crease. Matching is a defining rule of EZPR. The Riemann-hypothesis objective additionally requires a correspondence covering every nontrivial zeta zero and preserving its real part; that correspondence is not established in the archived proof.

I give an exact readout of the zeros of the Riemann zeta function on its critical line, preserving positions and multiplicities. The continued logarithmic derivative recovers the full nontrivial zero family as poles at their original positions. Under the guarantees reported in the archived rational contour review, exactly one simple zero lies in a specified rectangle; reflection symmetry fixes that zero on the critical line. The exact contour certificate and executable verifier remain pending retrieval. For every original nontrivial zero $\rho$, full recovery retains $(Y, d)$, where $Y = \operatorname{Im} \rho$ and $d = \operatorname{Re} \rho - ½$. Recovery from $Y$ alone holds if and only if $d = 0$ for every such zero. The established closure concerns admitted prime events and the signed critical-line readout.

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# A. Introduction

Riemann's 1859 paper connects prime counting to the zeros of the zeta function, beginning with Euler's product over primes [6]. Bombieri's exposition explains the hypothesis and why those zeros matter for the distribution of primes [7]. That connection motivates this investigation. The results below concern a separately defined coordinate model; the shared appearance of one-half does not identify its events with zeta zeros.

I named this construction Euler-Zeno in Prime Relatives (EZPR): a nod to Euler's primes and Zeno's half-steps. The construction and the claims below are mine.

The value is the same. Its address is different.

That distinction is where this construction starts. I can move a reference without changing its value. I can also give two different numbers the same location without making them the same number. To keep those statements straight, I need the number, its address and the frame that makes the address meaningful.

First, remove zero and one from the candidate family. Then remove the composites. Nine leaves because three times three is nine. Seven stays. The surviving labels are the primes. A number can leave the active comparison while its record stays available.

Now the placement question: where does a surviving number's comparison occur? I give the event an identified parent segment and read its position from both ends. The rule is that the two relative readings match. Equal progress from opposite ends of the same span puts the event halfway between them. The proof is short. Keeping track of what the proof actually uses takes more care.

Selection tells me which numbers remain. Matching tells me where their events sit. The first does not silently supply the second. In this paper, matching is an explicit rule of the construction.

Two, three and seven can therefore share a midpoint and still be two, three and seven. A coordinate is not an identity shredder. Their labels and histories stay with their records. If their parent spans differ, their actual locations can differ too: halfway along one span is not necessarily the same address as halfway along another.

This is where I use infinite addressing. I define the whole family, take an arbitrary member and prove the relation for it. The argument then covers every member. I do not need to plot each one to state that result. That is a universal argument, not a claim that a machine has finished an infinite run.

The research began with the Riemann hypothesis, which states that every nontrivial zeta zero has real part one-half [4, §25.10]. Sections C–H establish the coordinate construction. Sections I–L then use the common frame to derive the critical-line readout, recover original zero positions through the prime-power transform, and certify placement in one rectangle.

The order matters: identify the record, select the prime, place its matching event, then follow it through a change of frame and another generation. At the end, recover what was carried. Each step

has to leave the next one something it can actually use. The companion README gives the exact Lean coverage and reproduction details.

### a.1 Delivery mechanics and relative addresses

Follow the same participant through a change of address. Its identity stays with the record, and earlier records remain available for comparison. A new address does not require a new identity.

Definition — The continuing witness and its relative address.

A witness here is the identified participant being tracked. Let W denote that identity, n a nonnegative stage number, and $A_n$ its address on the chosen real coordinate line. For any position x on that line, define

$$r_n(x) = x - A_n \quad \text{EQ01}$$

Consequently,

$$r_n(A_n) = 0 \quad \text{EQ02}$$

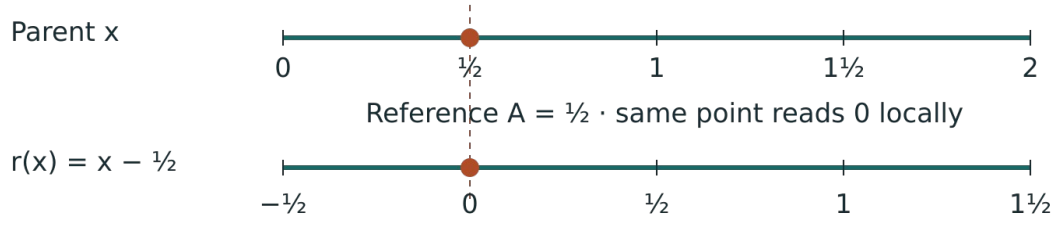
This is a translation of the coordinate origin [2, Chapter 2]: the witness is at zero relative to its own address. The zero is a reference reading, not a claim that the witness has disappeared or that a zeta function vanishes.

For example, at address $A_0=3$ the reading is $r_0(x)=x-3$. At address $A_1=8$ it is $r_1(x)=x-8$. The witness reads zero at each of those addresses. The addresses differ; the tracked identity W does not. These chosen addresses illustrate the definition; the formula does not generate the next address.

Figure 1 compares two readings of the same locations: the parent address and the address relative to one-half. Changing the reference does not move the point.

FIGURE 1

**Relative coordinates in half-unit steps**




A: reference address. r(x): relative reading. Dashed line aligns the same location in both charts.

EQ01–EQ02. Set A = ½. Parent readings 0, ½, 1, 1½, 2 become −½, 0, ½, 1, 1½. Only the origin changes; half-unit spacing and the underlying locations remain unchanged.

The retained stack means the earlier stage records remain available alongside the current record. For clarity, write that record sequence as

$$H_n = ((W, A_0), (W, A_1), ..., (W, A_n)) \quad \text{EQ03}$$

The stack keeps the identity and each recorded address together. Reading zero locally leaves those earlier addresses intact. This is the information represented by the example; the notation specifies no storage mechanism or additional inherited relationships.

Keeping a stack is useful only if I can get the required information back. That is the recovery question. Theorem 6 states the test precisely: recover after carrying, and obtain the inherited state. An implementation must satisfy that test to claim exact recovery.

## B. Definitions and retained state

Set operations and indexed families use standard notation [3, §§1.5, 1.8]. The records and admission rule below are definitions of the present model.

The definitions below separate the number being classified, the record that identifies it, and the position assigned to an admitted comparison event.

### b.1 Candidates and cancellation

Let $N_0$ denote the nonnegative integers:

$$N_0 = \{0, 1, 2, \ldots\}$$

A proper factorization means $n = u \times v$, where u and v are integers at least two. Define the composite set C and the surviving family S together:

$$C = \{u \times v : u, v \in N_0, u \geq 2, v \geq 2\}$$
$$S = N_0 \setminus (\{0, 1\} \cup C) \quad \text{EQ04}$$

Thus S is the surviving candidate family. Theorem 1 proves that S is exactly the primes; this equivalence is not needed as an assumption in its definition.

Cancellation excludes a candidate from this active family. It need not erase its retained record. Excluding the numerical candidates zero and one also does not remove zero and unit reference roles from the coordinate construction.

### b.2 Records, parents and positions

A label identifies a number. A record keeps that label together with the information required to interpret its event. For each $p \in S$, the proposed placement model supplies a record containing p, participant identities, parent identity $K_p$, endpoints $a_p<b_p$, event location $x_p$ and relevant history.

The parent span is the segment between those endpoints. Its length is $L_p=b_p-a_p>0$. All event locations belong to the declared affine X line. There is no independent Y component in this model's location type; this is a model definition, not a reduction theorem for another system.

Distinct labels may share a location. Equal endpoint values do not by themselves identify participants or their histories.

### b.3 Relative readings, incidence and matching

For the same event and identified parent, define

$$t_p = (x_p - a_p)/L_p,\ u_p = (b_p - x_p)/L_p \quad \text{EQ05}$$

These are directed parent-relative coordinates. For an event inside the parent segment, they are the fractions of the span traversed from its two endpoints. The positive length makes both expressions well-defined. If participants begin in different charts, a supplied relative-coordinate map must first identify the common event and parent being compared.

Incidence is the identity $t_p+u_p=1$, obtained directly from these definitions. Matching is the additional admission rule $t_p=u_p$. A matching record satisfies both equations and the location relation $x_p=a_p+L_p\ t_p$. Theorem 2 derives its unique midpoint location.

The definitions do not derive matching from primality, from available perspectives, or from the equality of reference labels. The whole-family theorem explicitly requires a matching record for each survivor.

### b.4 Retention and terminology

Here “crease” means the identified parent midpoint. “Projector” names a participant role in the source construction; the affine equations below represent a comparison and do not independently establish native projector behavior. “Closure” means that the declared conditions imply the stated conclusions throughout the admitted family.

Retention means keeping enough information to recover the specified inherited state. Theorem 6 states the exact recovery condition; storing a history label alone is not that condition.

The revised terminology distinguishes perspective, connecting relation, comparison and reception. Perspective alone does not imply a receiver, reflection, doubled scale or new coordinate axis. Value, Perception and Action describe respectively the retained state, its admitted comparison and the committed operation; this usage introduces no physical timing schedule.

# C. Prime Survivor Equivalence Theorem

Start with the whole numbers. Remove zero and one, then every number made by multiplying two whole numbers greater than one. Nine goes because three times three is nine; seven and two stay. Every prime survives, and every survivor is prime. The proof below establishes both directions.

### Statement and proof

We use the standard definition of prime and composite integers [1, Definition 1.1.3].

For $n \in \mathbb{N}_0$, $n \in S$ if and only if n is prime.

$$n \in S \Leftrightarrow n \text{ is prime}$$

Proof. Membership supplies $n \geq 2$ and excludes every factorization $n=uv$ with both factors at least two. Thus n has no proper divisor d with $2 \leq d<n$: such a divisor would give $n=dv$ and $v \geq 2$. Conversely, if n is prime, $n \geq 2$ and any such factorization would provide a proper divisor u, because $v \geq 2$ forces $u<n$. Hence no proper factorization exists and $n \in S$. □

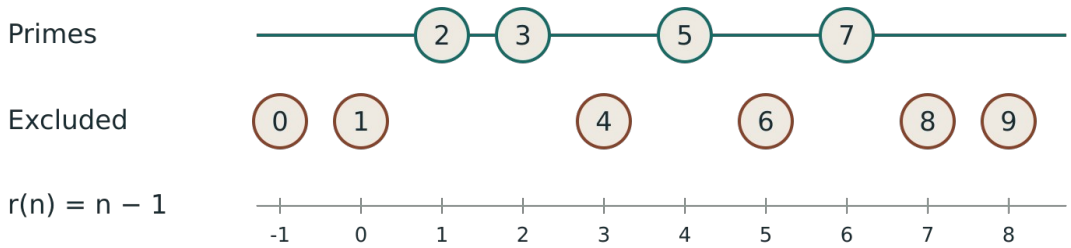
The statement classifies the whole family by the standard prime/composite distinction [1, §1.1.1]. It supplies no midpoint-placement theorem.

This selection classifies labels. It does not say that a geometric point labeled 9 cannot be placed at a midpoint.

Figure 2 separates the surviving prime labels from the excluded labels. The diagram illustrates the selection rule; it does not supply placement.

FIGURE 2

### Separate the family from the exclusions




Circled labels link to their definitions. The upper line passes through prime positions; circle interiors keep numerals readable.

Theorem 1. Primes remain on the upper row; excluded labels sit below. The aligned scale $r(n) = n - 1$ uses reference 1: its local reading is 0. This coordinate change does not change primality.

# D. Unique Matched Midpoint Theorem

The coordinate representation is an instance of affine and barycentric coordinates [2, §§2.2, 2.6]. The equal-progress condition is imposed separately by this model.

The quotient definitions imply incidence:

$$t_p + u_p = 1 \quad \text{EQ06}$$

In this paper's declared model, the EZPR admission rule is equal relative progress:

$$t_p = u_p \quad \text{EQ07}$$

The first equation follows from a shared parent and event. The second specifies which event is admitted. Neither primality nor two available perspectives alone implies the second equation.

The same relations can be expressed by inward parameterized rays

$$E_p(t) = a_p + L_p\, t,\ T_p(u) = b_p - L_p\, u,\ t, u \geq 0 \quad \text{EQ08}$$

Their incidence is equivalent to t+u=1. Their images overlap on the parent interval; unique placement arises only after matching is imposed. The parameters are coordinates, not elapsed time.

Take a segment from zero to ten. Five is equally far from both ends; four is four units from one end and six from the other. The matching rule requires equal readings from the same event and parent span. Only the midpoint satisfies it. This proves the consequence of matching, not that primality itself creates matching.

### Statement and proof

For real endpoints a<b and real unknowns t,u,x, the incidence, matching and location equations

$$t + u = 1,\ t = u,\ x = a + (b - a)t \quad \text{EQ09}$$

have exactly one solution: t=u=1/2 and x=(a+b)/2.

$$t = u = \tfrac{1}{2} \quad \text{and} \quad x = (a + b)/2$$

Proof. Substituting u=t gives 2t=1, so t=u=1/2. The location equation then gives x=(a+b)/2. Conversely those values satisfy every equation, establishing existence as well as uniqueness. □

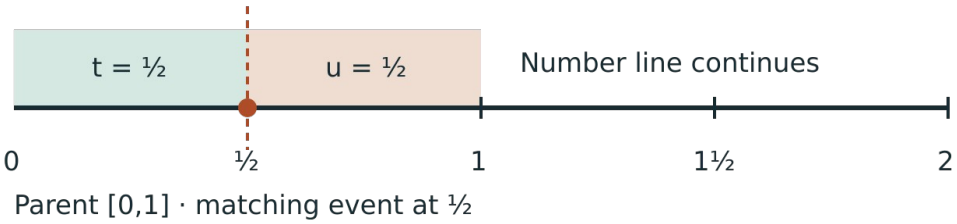
Take the surviving label 7 with parent endpoints 0 and 2. Its matched event has X=1 and normalized progress 1/2 from either side. The same label with parent endpoints 10 and 14 has

X=12 and again normalized progress 1/2. The label and normalized relation persist; the global location changes with the parent.

Figure 3 shows the matching event on its identified parent span. The extension beyond that span does not change which endpoints define the midpoint.

FIGURE 3

**Matching fixes the midpoint**




t, u: fractions read from opposite ends. Dashed line: midpoint marker. Dot: event.

Placement (Theorems 2–3). Incidence gives t + u = 1; the supplied matching rule t = u forces x = ½ on the normalized parent [0,1]. The parent remains [0,1]; 1½ and 2 show the continuation beyond it.

# E. Local Prime-Family Closure Theorem

Combine the first two results: select the prime labels, then equip each with a matching record. Each event lies at its own parent midpoint. Several distinct primes can share a location without losing their identities. Different parent spans need not have the same midpoint.

### Statement and proof

Suppose every $p \in S$ is equipped with real endpoints $a_p<b_p$ and real readings $t_p,u_p,x_p$ satisfying $t_p+u_p=1$, $t_p=u_p$ and $x_p=a_p+(b_p-a_p)t_p$. Every prime record then lies at its identified parent's midpoint. No such record lies strictly to either side of that midpoint.

$$\forall p \in S: \quad x_p = (a_p + b_p)/2$$

Proof. Theorem 1 identifies S with the primes. Apply Theorem 2 to an arbitrary member. Either strict inequality contradicts the midpoint equality. Because the member was arbitrary, the conclusion holds universally without enumeration. □

Corollary — Shared parent, common X.

If all records have the same endpoints a,b, all matched locations equal (a+b)/2. More generally, equal parent midpoints suffice. Equal normalized progress alone is insufficient when the midpoints differ.

Define the normalized representation $Z=\{(p,1/2):p \in S\}$. Projection $(p,1/2)\mapsto p$ is bijective onto S: its inverse is $p\mapsto(p,1/2)$, and composing these maps in either order returns the starting element. Projection onto 1/2 alone is not injective: the distinct labels 2 and 3 already provide a counterexample. The coordinate tells us where; the retained record tells us what arrived. Losing the label is not a shortcut to remembering it.

# F. Affine Preservation Theorem

Affine maps preserve affine combinations [2, §2.7, Definition 2.6]. The midpoint case uses two weights, each equal to one-half; the proof below specializes that standard fact.

Shift or uniformly rescale both endpoints and their event together. On a segment from zero to ten, the midpoint is five. Double all three coordinates and add three: the endpoints become three and twenty-three, and the midpoint becomes thirteen. The same relationship survives. A zero scale is excluded because it collapses the span; a negative scale reverses orientation.

### Statement and proof

For $f(x)=\alpha x+\beta$ with $\alpha \neq 0$, midpoint membership is preserved when the complete event and parent are transported together.

$$x = (a + b)/2 \Rightarrow f(x) = (f(a) + f(b))/2$$

Proof. If $x=(a+b)/2$, then $f(x)=(f(a)+f(b))/2$ by distribution. Also $f(b)-f(a)=\alpha(b-a) \neq 0$, so the normalized comparison remains defined. Set $a'=\min(f(a),f(b))$, $b'=\max(f(a),f(b))$, and $L'=b'-a'=|\alpha| L>0$. Use $t'=(f(x)-a')/L'$ and $u'=(b'-f(x))/L'$. When $\alpha<0$ the endpoint order reverses, so $t'=u$ and $u'=t$; when $\alpha>0$, $t'=t$ and $u'=u$. Thus matching is preserved in either case. Participant identities remain in their records independently of endpoint order. □

# G. Recursive Preservation Theorem

The proof uses ordinary mathematical induction [3, §10.1] and the definition of an indexed union [3, §1.8].

Start with matching records. At each next stage, preserve the inherited matches and require every new record to match. Every resulting stage then satisfies the rule. Any record in the union entered at some finite stage, so it satisfies the rule too. This does not establish that every prime eventually enters.

### Statement and proof

Let $G_0$ consist of matched events. Suppose each $G_{n+1}$ consists of inherited events whose parent-relative matching is preserved and new events satisfying incidence, matching and location. Then all events at every finite stage match.

$$\forall n \in N_0, \forall q \in G_n: q \text{ satisfies matching}$$

Proof. The base is given. At the successor stage, inherited events match by preservation and new events by Theorem 2. Induction completes the proof. Every member of the union of finite stages also matches because it belongs to some stage. □

Every record that enters stays matched under these rules. To claim every prime is represented, I must also show that each prime enters. Induction keeps the records in order; it does not fill the guest list.

# H. Recoverable Carry is Injective Theorem

The argument uses function composition and injectivity [3, §§12.2, 12.4–12.5]. Exact recovery is the left-inverse condition; the proof below derives injectivity directly.

If carrying could turn two different inherited states into the same encoded state, one recovery operation would have to return two different originals from that same input. Exact recovery rules this out: identical encodings must have identical originals. New experience may still be added alongside the inherited component.

### Statement and proof

Let $H_{set}$ be the admitted inherited states and $B_{set}$ the encoded states. If functions $K:H_{set}\to B_{set}$ and $R:K(H_{set})\to H_{set}$ satisfy $R(K(H))=H$ for every $H \in H_{set}$, then K is injective.

$$R \circ K = \mathrm{id} \Rightarrow K \text{ is injective}$$

Proof. $K(H)=K(J)$ implies $H=R(K(H))=R(K(J))=J$. □

An enlarged successor can contain additional experience. The identity applies to its recoverable inherited component, not to equality of the entire enlarged state. Historical admission and current read validity also remain distinct.

### h.1 Worked example — Keep the same record

Start with seven and keep nine beside it as a control. Nine is excluded because $9=3\cdot3$. Seven survives. This selects the label; it has not yet placed an event.

Give seven an identified parent P with endpoints 0 and 10. Its readings are $t=x/10$ and $u=(10-x)/10$. Require matching $t=u$. Solving gives $x=5$ and $t=u=1/2$. The parent and matching rule are supplied; the midpoint is calculated.

Keep the record $Q_0$=(label 7, parent P, endpoints (0,10), location 5). Its participant identities are retained as additional fixed fields. Another prime can occupy the same location without replacing

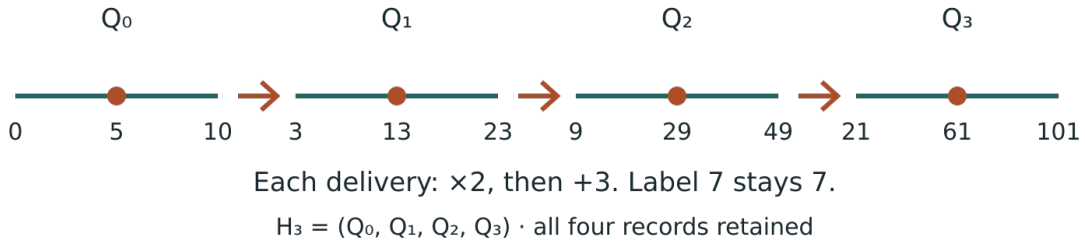
label 7. Nine's exclusion is arithmetic: nothing prevents an unrelated geometric point labeled nine from sitting at five.

Now apply f(x)=2x+3 to the entire parent and event. The new endpoints are 3 and 23; the event is at 13. The two relative readings are (13−3)/20=1/2 and (23−13)/20=1/2. Retain $Q_0$ and the transformation alongside the new record $Q_1$. The ordered history is $H_1=(Q_0,Q_1)$. We changed the address, not the label.

Figure 4 follows the same label through four deliveries. Each new record changes the parent and event addresses while retaining the earlier records.

FIGURE 4

**Delivery mechanics through $Q_3$**




$Q_0$–$Q_3$: successive records. Dots: event locations. Each span uses its own display scale.

Theorems 4–5. Apply $x \mapsto 2x + 3$ at each step to both endpoints and the event. Midpoints read 5, 13, 29, 61; every event remains halfway within its own parent. Label 7 is retained throughout.

The relative-address formula reads this change locally: $r_0(x)=x-5$ and $r_1(x)=x-13$. Each event reads zero at its own address. The retained stack keeps both addresses and their parent records. Local zero is not the midpoint fraction: zero describes recentering at the event, while one-half describes progress through the parent.

For successive stages, repeatedly apply the same affine transformation to the preceding parent and event, retaining the earlier records. Theorem 4 preserves matching at each step; Theorem 5 therefore preserves it at every finite stage. This example follows seven throughout. It does not claim to generate every prime.

Finally, make recovery explicit for this illustrative record scheme. Carry an inherited stack H with new experience e by K(H)=(H,e), choosing e as a fixed or specified function of H. Define R(H,e)=H. Then R(K(H))=H exactly. This simple retained-pair encoding witnesses Theorem 6; it is not a claim that a native implementation uses this storage format.

That is the complete example: select the label, supply the comparison rule, calculate its place, move the whole parent, retain the earlier records and recover them. Each operation has a stated job. None borrows a conclusion from the next one.

# I. Common-frame prime crease

Theorems 1–3 establish prime admission and parent-normalized placement. Theorems 4–6 preserve the construction under affine transport, recursive admission, and exact recovery. I now fix a common frame so that the parent midpoint can serve as the reference for the zeta readout.

**Worked record.** Admit prime label 2 in the unit parent. The event reads one-half from either end. The record still says 2 after it takes the turn. A second parent could have a different absolute midpoint even though its normalized reading is also one-half. A common crease uses a common chart and coincident parent midpoints in that chart. [12]

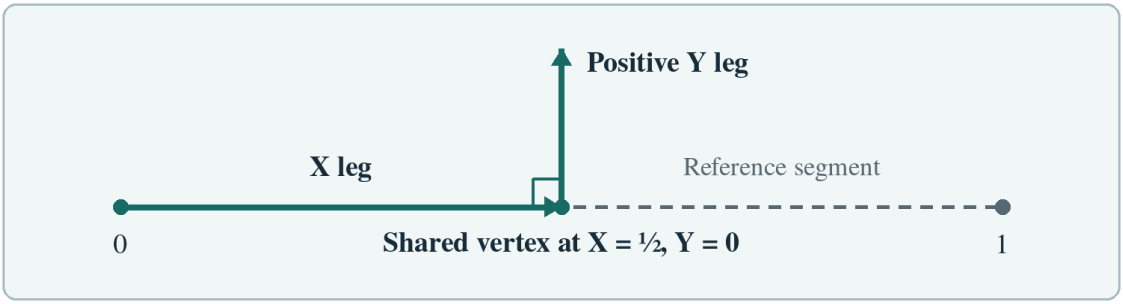



*Figure 5. The unit reference dyad and the positive bent path. The unused half of the reference segment remains visible in gray.*

**Theorem 7 (The turned leg and its signed chart).** In the common Euclidean unit frame the reference endpoints are 0 and 1. Their perpendicular bisector is $X=1/2$. Put $C(Y)=1/2+iY$ for all real $Y$. Define $B(t)=t$ for $0\le t<1/2$ and $B(t)=C(t-1/2)$ for $t\ge 1/2$.

**Proof.** A point on the perpendicular bisector is equally distant from the unit endpoints. Equating squared distances gives

$$X^2 + Y^2 = (1 - X)^2 + Y^2 \iff X = ½$$

Both legs meet at $B(1/2)=C(0)$. On the turned leg $B(1/2+Y)=C(Y)$ for $Y\ge 0$. The signed chart also permits negative heights. Its inverse is the imaginary part. On the path, the distance parameter is $X$ before the turn and $1/2+Y$ after it. [13, 17]

# J. Exact critical-line readout

**Theorem 8 (Exact zero readout).** Put $F(Y)=\zeta(C(Y))$. Its zeros correspond bijectively to the zeta zeros on the crease, with their original positions and multiplicities.

**Proof.** Each prime contributes $p^{-X}e^{-iY\log p}$ at $s=X+iY$. For $X>1$, the Euler product of $(1-p^{-s})^{-1}$ equals $\zeta(s)$. Evaluate its analytic continuation on the crease. Then $F(Y)=0$ exactly when $C(Y)$ is a crease zero, and $Y=\Im C(Y)$ is the unique inverse. At a zero $\rho=C(Y_0)$ of order $m$, write $\zeta(s)=(s-\rho)^m g(s)$ locally with $g(\rho)\neq 0$. In the holomorphic extension in $Y$,

$$F(Y) = i^m (Y - Y_0)^m g(C(Y)). \qquad (1)$$

The cofactor is nonzero at $Y_0$, so the order is unchanged. This is local closure for the entire signed crease. [13, 17] See [4, §§25.2 and 27.4] for the Euler product and continuation.

## K. Prime-power transform and pole recovery

**Theorem 9 (Original zero recovery).** In the open critical strip, the poles of the continued aggregate prime-power transform are precisely the nontrivial zeta zeros. Each pole is simple and its residue equals minus the zero's multiplicity.

**Proof.** For $\Re\, s > 1$, logarithmic differentiation of the absolutely convergent Euler product gives

$$D(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} = -\frac{\zeta'(s)}{\zeta(s)}. \qquad (2)$$

The quotient supplies meromorphic continuation. Let ρ be an arbitrary zero in the open critical strip. Zeta is holomorphic there and is not identically zero: its continuation is analytic on the connected domain $\mathbb{C} \setminus \{1\}$ and $\zeta(2) \neq 0$. The identity theorem therefore makes ρ an isolated zero of finite order $m \geq 1$. The local order-of-vanishing theorem gives $\zeta(s) = (s - \rho)^m g(s)$, with g holomorphic and $g(\rho) \neq 0$. Shrink the neighborhood so g stays nonzero. Differentiation yields $D(s) = -m/(s - \rho) - g'(s)/g(s)$. The second term is holomorphic. At every nonzero point in the strip, the quotient is holomorphic. Thus the poles recover precisely the original zeros, and their negative residues recover multiplicities. This argument applies to arbitrary strip zeros, independently of their real coordinates. [4, 17]

For each original zero $\rho$, retain its height $Y = \Im\, \rho$ and its real offset $d = \Re\, \rho - 1/2$. The crease point at the same height is $C(Y) = 1/2 + iY$. Subtracting gives $\rho - C(Y) = d$, and adding the offset back recovers $C(Y) + d = \rho$.

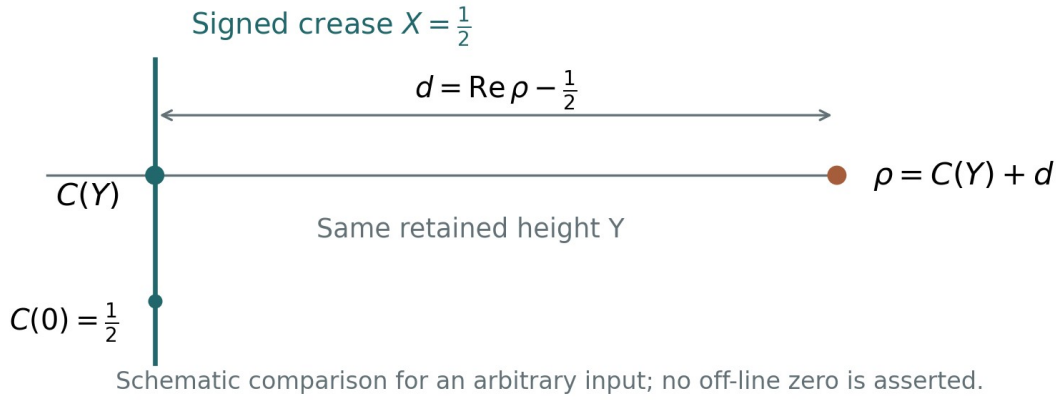



*Figure 6. The original point and the crease point at the same height. The full address retains d; the Y-only return equals the original point exactly when d = 0. The diagram is schematic and asserts no off-line zeta zero.*

**Corollary (Full-strip equivalence).** Exact recovery by $Y$ alone for every original nontrivial zero is equivalent to $\Re\,\rho = 1/2$ for every such zero.

**Proof.** For an arbitrary zero the difference between the two returns is $\rho - C(\Im\,\rho) = d$. Thus the returns agree exactly when $d = 0$. Quantify over the full family. The statement proves the equivalence. The universal assertion is open in the proof record. [15, 17]

## L. Certified placement in a bounded rectangle

Proposition (Placement under the reported contour guarantees). Let U be the rectangle $0.495 \le X \le 0.505$, $14.1297 \le Y \le 14.1397$. Assume the reported 128 rational boundary enclosures rigorously exclude zero and establish winding number one. Zeta is holomorphic throughout U, so the argument principle counts one zero with total multiplicity one. That zero is simple. The rectangle is invariant under J(s) = 1 − conjugate(s), and the functional equation and conjugation symmetry preserve strip zeros under J. The image of the unique zero must be that same zero. Thus ρ = J(ρ), and Re ρ = ½. The conclusion depends on the stated contour guarantees. [4, §§25.4 and 25.10; 18]

Archived replay report. The source review reports exact replay of 128 boundary enclosures in the same implementation. The exact certificate and executable verifier remain pending retrieval, so this revision records no independent replay or new numerical certification. [16]

Reported finite controls. The November 3 record reports 290 finite control passes and a residual of approximately $1.826 \times 10^{-12}$ near ½ + 14.134725141737i. These are historical reported observations. A small residual alone does not prove a zero; the bounded zero count follows from the stated contour guarantees. [16]

## M. Summary of results

The construction has three jobs: select the numbers, place their events and preserve the records. Each theorem settles one part of that work.

**Selection.** Removing zero, one and proper products leaves exactly the primes (Section C).

**Placement.** When two readings of the same event use the same parent span and satisfy the matching rule, the event is uniquely at that span's midpoint (Section D). Applying this result to each supplied matching record gives local prime-family closure (Section E). Selection alone does not supply matching.

**Preservation.** A common affine change preserves midpoint placement (Section F). Valid initial records and a validity-preserving recursion preserve it at every finite stage (Section G). Exact recovery makes the carry map injective on inherited states (Section H).

Together, Theorems 1–6 close the declared EZPR model under its admission and preservation premises. The common frame identifies its crease at one-half (Section I). The exact readout preserves crease zeros and multiplicities (Section J). The prime-power transform recovers all original strip zeros at their original positions (Section K). Section L derives bounded placement from the reported contour guarantees and states their reproduction dependency. Full-strip recovery from Y alone remains equivalent to universal placement.

## N. Discussion and conclusion

Two comparisons place this result in the wider literature. Lagarias proves that a sequence of inequalities involving harmonic numbers and sums of divisors is equivalent to the Riemann hypothesis [9, Theorem 1.1]. This demonstrates that a formulation using integers can address the hypothesis, provided the equivalence itself is proved. EZPR's classification and midpoint theorems do not yet supply such an equivalence.

Platt and Trudgian rigorously verify the location of zeta zeros up to height $3 \times 10^{12}$, using interval arithmetic [10]. Their result concerns a bounded region of the zeta function. Here, universal quantification covers every admitted EZPR record, but the objects quantified over are model records. A large verified region, an entire defined family, and an equivalence with the zeta-zero problem are distinct mathematical achievements.

What did the construction deliver? Cancellation leaves exactly the primes. Matching puts each admitted event halfway along its own parent. When records share a parent, they share that location. Their identities stay with them.

Now move it. Shift the whole parent or change its scale, and the event stays halfway. Carry the records through another generation under the preservation rules, and the match survives. Exact recovery answers the last question: can I get the inherited information back? If I can, two different inherited states cannot have been merged along the way.

That is local closure in EZPR. I supply equal-progress matching as a construction rule; the midpoint follows from it, incidence and location. Primality determines the family. Matching determines placement. The companion README traces the relationship to the earlier research constructions.

The native constructor must still supply the declared matching records. For the zeta zeros, Section K gives a full address consisting of the height and the real offset. Recovering every original zero from the height alone is equivalent to proving that each offset vanishes. Section L establishes this placement for the zero inside the certified rectangle.

Return to the opening line. The value is the same. Its address is different. We can now say exactly what survives that change: identity, matched placement under the stated rules, and inherited information when recovery is exact. The record has moved. We can still account for what arrived.

# O. Methodological note

I supplied the construction, terminology corrections and research direction. AI assistance helped organize formal statements, proofs, source reconciliation and the Lean artifact. Source declarations, machine-checked consequences and open correspondence obligations are distinguished throughout. My writing preferences guide the explanation; the proofs carry the mathematical claims.

# P. Cross-examination and disproof tests

### p.1 Proof cross-examination

The questions belong to the proof record. Some are tests of the coordinate model. Others check the analytic return. I keep their answers with the step they challenged.

### q.1 Which labels survive?

- The surviving labels are exactly the primes. Excluding zero, one, and every product of two integers at least two removes precisely the nonprime candidates. [Theorem 1, ref. 12]

### q.2 Where does a matched prime dyad sit on X?

- Within its own parent span, the matched event sits halfway between the endpoints. Its normalized coordinate is ½.
- The absolute coordinate is also ½ when the parent midpoint is ½ in the shared X chart. [Theorems 2 and 7, ref. 12]

### q.3 What fixes the bend and its direction?

- In the unit frame, the points equally distant from 0 and 1 form the perpendicular bisector X = ½. The turned leg follows that line. [Theorem 7]

**q.4 Where is the shared vertex?**

- The vertex is the same point in both descriptions. Its coordinate on the unit X segment is ½. Its signed Y coordinate is 0. [Theorem 7]

**q.5 Does the join make a grid or add a free coordinate?**

- One distance parameter follows the path from the X leg through the bend and along the turned leg. Once on the turned leg, only Y varies. [Theorem 7]

**q.6 Can a returned Y-line zero be placed elsewhere?**

- Each zero returned by the Y readout maps to exactly one point, ½ + iY. On $Y \geq 0$, the positive bent path returns the same point as the signed chart. [Theorems 7–8]

**q.7 Can the crease readout lose, displace, or miscount a crease zero?**

- Taking the imaginary part of the returned point recovers its Y address.
- The affine substitution contributes a nonzero factor to the local zero expansion, so the order of vanishing stays the same. [Theorem 8]

**q.8 Do the prime modes share Y?**

- Yes. Every prime uses the same height Y. The factor $\exp(-iY \log p)$ gives each prime its own phase rate $\log p$. [Theorem 8]

**q.9 Does the prime spectrum reproduce classical zeta there?**

- The Euler product equals $\zeta(s)$ where $\mathrm{Re}\, s > 1$. Continue that function analytically, then evaluate it at ½ + iY. [Theorem 8, ref. 4]

**q.10 Does the aggregate recover original zeros and their multiplicities?**

- In the open critical strip, the continued transform has a pole at each original nontrivial zero and nowhere else.
- The pole location gives the original zero. The negative of its residue gives the zero's multiplicity. [Theorem 9]

**q.11 Does infinite addressing reach all original nontrivial zeros?**

- Yes. The full address stores the height Y and the real offset d. Adding d to the crease point at height Y recovers the original zero. [Theorem 9 and Section K]

**q.12 What does the offset measure, and where does the comparison cancel?**

- The offset d is the signed difference between the original real coordinate and ½. Compare both points at the same Y.
- The comparison gives zero precisely when the original point is already on the crease. [Section K, ref. 15]

### q.13 What does the same-height comparison retain?

- The height $Y$ identifies the crease point. The offset $d$ retains the real displacement needed to recover the original zero. [Section K]

### q.14 Can rotation of the whole output create a zero?

- No. Multiplying the output by a unit complex phase leaves its magnitude unchanged, so a nonzero value stays nonzero. [t.5]

### q.15 Does signed mirror balance certify each address?

- No. Two opposite offsets can sum to zero while each is nonzero. A signed sum alone cannot establish placement at each address. [t.6]

### q.16 What did the contour prove, and what checks passed?

- Under the reported contour guarantees, the argument principle counts exactly one simple zero in the specified rectangle. Reflection symmetry puts it at X = ½. The exact certificate and verifier remain pending retrieval. [Section L, ref. 16]
- The record reports an exact replay of 128 boundary enclosures in the same implementation and 290 separate finite control passes. [Section L, ref. 16]

### q.17 Is counting poles and line zeros at every height a new objection?

- With the same signed height window and multiplicities, every crease zero is already included in the full count.
- Equality at every cutoff would exclude any extra off-crease zero. This is the counting form of the same full-family condition. [t.8]

### q.18 What does the readout establish outside its domain?

- The readout identifies every zeta zero on its own crease. The full transform separately recovers each original zero's height and offset.
- Recovery from Y alone agrees with full recovery exactly when the offset vanishes at that address. [Theorems 8–9 and Section K]

## p.2 Disproof attempts and outcomes

### t.1 Break prime admission or move the midpoint

Try to admit a nonprime survivor or place an admitted matched event away from its parent midpoint. Theorem 1 characterizes the survivors exactly. Theorems 2–3 force each admitted event to its identified parent midpoint. Absolute X = ½ additionally requires the common-frame hypothesis in Section I. The challenge is ruled out within those stated premises.

### t.2 Break the bend or introduce a second free coordinate

Try to shift the relative origin away from the vertex or give one point on the turned line two altitude addresses. **Ruled out.** $C(0) = v$ and $\mathrm{Im}\, C(Y) = Y$. The two legs share the vertex, and the turned line has one free coordinate.

### t.3 Lose a zero in either recovery

Try to lose a crease zero, change its multiplicity, or lose an original nontrivial zero in the aggregate. **Ruled out in each stated domain.** The Y inverse and local factorization preserve the crease zeros. The poles and residues of D, together with the full relative address, preserve the original zeros and their multiplicities.

### t.4 Break prime-spectrum agreement

Compare the analytically continued prime formula with the classical zeta readout on the crease. **Exact agreement.** The Euler product gives the same function on $X > 1$, and uniqueness of analytic continuation carries that equality to the crease.

### t.5 Manufacture cancellation by rotating the output

Rotate the whole complex output and test whether a nonzero value can become zero. **Ruled out.** For real $\theta$, $|\exp(i\theta)\, \zeta(s)|^2 = |\zeta(s)|^2$.

### t.6 Hide displacement through mirror balance

Test whether opposite offsets can cancel in the aggregate while each remains nonzero. **The control defeats that shortcut.** A pair $d, -d$ has signed sum zero for every $d$. Signed pair balance cannot certify cancellation at each original address.

### t.7 Challenge the bounded certificate and finite controls

Section L derives the bounded conclusion from reported contour guarantees and distinguishes the historical finite controls. The exact certificate and executable verifier remain pending retrieval; this revision records no fresh replay. [16]

### t.8 Challenge by counting at every height

Compare the full pole count and the crease-zero count, including multiplicities. For each $T > 0$ use the signed open window $-T < \mathrm{Im}\, \rho < T$ in the full strip and $-T < Y < T$ on the crease. Both counts exclude the endpoints. **Duplicate of the full RH claim.** Since crease zeros are already included with matching multiplicities, equality at every cutoff is equivalent to equality of the two families.

# Index of definitions

Definitions below specify the domains, conditions, and meanings used in the proofs. Numerical entries are worked instances of those definitions. Native terminology is distinguished from the mathematical structure actually defined in this paper. For $n \geq 1$, $D(n)$ denotes the positive divisors of $n$; $S$ and $C$ denote the survivor and composite sets defined in Section b.1.

**0 side of the 0 < 1 dyad** — For a parent $a<b$, $t(x) = (x-a)/(b-a)$ gives $t(a) = 0$. Separately, a chosen reference $A$ has local coordinate $r_A(A) = 0$ under $r_A(x) = x-A$. These zeros coincide only when $A = a$; a midpoint reference has $t(A) = ½$ while $r_A(A) = 0$. Neither reference convention makes integer 0 a prime. Sections a.1 and b.3.

**1 side of the 0 < 1 dyad** — The one endpoint sets the unit span relative to zero. Excluding integer 1 from the prime family does not remove the unit used in comparison. $L = b - a > 0$; $t(x) = (x - a)/L$; $u(x) = (b - x)/L$; $t + u = 1$. EQ05–EQ06: parent-relative pair. At a: $(t,u) = (0,1)$; at b: $(1,0)$. This is the paper's dyad readout, not a replacement native definition.

**2 · prime label and denominator** — For an integer $n \geq 1$ define $D(n) = \{d \in \mathbb{N} : d \text{ divides } n\}$, where $\mathbb{N} = \{1,2,\ldots\}$. Then $D(2) = \{1,2\}$, so 2 is prime. Separately define $d_k = k/2$ for $k \in N_0$; this gives 0,½,1,1½,2,… as half-unit coordinates. The denominator specifies a coordinate spacing, not the prime-selection predicate. Sections b.1 and C.

**3 · a prime and a comparison value** — 3 is prime. Two occurrences of 3 may enter a multiplicative comparison: $3 \times 3 = 9$. Their equal values do not require the two records to be identical. $D(3) = \{1,3\}$; $3 \times 3 = 9$. Theorem 1: prime label and proper-product witness.

**4 · composite** — $2 \times 2 = 4$. Excluded from the prime candidate family; its record may remain in history. $4 = 2 \times 2 \in C$; $4 \notin S$. EQ04.

**5 · prime** — No factorization into two integers greater than one. Retained in the prime candidate family. $D(5) = \{1,5\}$; $5 \in S$. Theorem 1.

**6 · composite** — $2 \times 3 = 6$. Excluded from the prime candidate family; its record may remain in history. $6 = 2 \times 3 \in C$; $6 \notin S$. EQ04.

**7 · prime** — No factorization into two integers greater than one. Retained in the prime candidate family. $D(7) = \{1,7\}$; $7 \in S$. Theorem 1.

**8 · composite** — $2 \times 4 = 8$. Excluded from the prime candidate family; its record may remain in history. $8 = 2 \times 4 \in C$; $8 \notin S$. EQ04.

**9 · composite** — $3 \times 3 = 9$. Excluded from the prime candidate family; its record may remain in history. $9 = 3 \times 3 \in C$; $9 \notin S$. EQ04.

**Affine transport** — Applying $f(x) = \alpha x+\beta$, $\alpha \neq 0$, to the event and both endpoints together. Section F, Theorem 4; standard mathematics. See [2, §2.7].

**Bent path and signed chart** — The nonnegative distance parameter traverses the X leg and then the positive Y leg. The signed crease chart includes both positive and negative heights. See Section I.

**Bijective / inverse** — A map $f:X\to Y$ is bijective if it is injective and $\forall y \in Y$ there exists $x \in X$ with $f(x) = y$. Its inverse $g:Y\to X$ satisfies $g\circ f = id_X$ and $f\circ g = id_Y$. A left inverse alone proves injectivity, not surjectivity onto an independently specified Y. Sections E and H. See [3, §12.5].

**Cancellation** — The set operation $S = N_0\setminus(\{0,1\} \cup C)$, where $C = \{uv : u,v \in N_0, u \geq 2, v \geq 2\}$. Thus a label is cancelled precisely when it lies in $\{0,1\} \cup C$. This operation acts on membership in S; it specifies no deletion operation on stored records. Sections b.1 and C.

**Candidate** — An element n of $N_0 = \{0,1,2,\ldots\}$, before application of the selection predicate. The candidate domain includes 0 and 1; membership in that domain does not imply primality. Section b.1.

Carry / decoder / exact recovery — Let $\mathscr{H}$ be the inherited-state domain and $\mathscr{E}$ the encoding space. An encoder $K:\mathscr{H}\to\mathscr{E}$ has a decoder $R:K(\mathscr{H})\to\mathscr{H}$. Exact recovery means $R(K(h)) = h$ for every $h \in \mathscr{H}$. Consequently K is injective. The illustrative encoding $K(h) = (h,e)$ retains the declared inherited component and makes no storage-efficiency claim. Section H.

**Composite / proper factorization** — A number expressible as uv with whole-number factors $u,v \geq 2$. C denotes that composite set. Section b.1; standard mathematics. See [1, §1.1.1].

Delivery mechanics — The continuing-witness example defined in Section a.1: $rn(x) = x-An$ and $rn(An) = 0$. W is the retained identity, An the stage address, and Hn the displayed sequence of earlier identity-and-address records. The formula recenters; it does not generate addresses.

Euler-Zeno in Prime Relatives (EZPR) — Euler-Zeno in Prime Relatives (EZPR) is the construction specified in Sections B–E: select S, associate retained records with labels $p \in S$, supply a parent span and two readings for each admitted event, and impose equal readings. Selection and matching are separate conditions. Theorems 1–6 in Sections C–H establish consequences of those conditions.

Event — The marked location $x \in \mathbb{R}$ in a specified parent record, together with its readings $t = (x-a)/(b-a)$ and $u = (b-x)/(b-a)$. For an event inside its parent, $a \leq x \leq b$ and $t,u \in [0,1]$. Matching additionally requires $t = u$. The term does not introduce a time variable or physical propagation law. Sections b.2–b.3.

**Frame / chart** — A coordinate chart assigns real addresses to the locations under consideration. This paper uses relative translation $r_A(x) = x-A$, $A \in \mathbb{R}$, and affine changes $f(x) = \alpha x+\beta$, $\alpha \neq 0$. Their inverses are $r_A^{-1}(y) = y+A$ and $f^{-1}(y) = (y-\beta)/\alpha$. A comparison across charts requires the specified connecting map; equal written numerals in different charts alone do not identify a location. Sections a.1 and F.

**Generation / recursive preservation** — Let $G_n$ be the record collection at stage $n \in N_0$. Assume every record in $G_0$ is valid, inherited records retain validity at the next stage, and every newly admitted record is valid. Induction then gives validity throughout every $G_n$. Here validity means satisfaction of the declared parent, incidence, and matching conditions. Section G.

**Incidence** — For $E(t) = a+Lt$ and $T(u) = b-Lu$, with $L = b-a>0$, incidence means $E(t) = T(u)$. Subtracting a and dividing by L gives $t+u = 1$. This equality identifies a common location but does not by itself require $t = u$. Section D.

**Induction** — Prove a starting stage and prove that validity at one stage implies validity at the next. The result then holds at every finite stage. Section G; standard proof method. See [3, §10.1].

**Infinite addressing** — The paper's term for specifying an entire indexed family and proving a universally quantified property: for an arbitrary $p \in S$, prove $P(p)$, hence $\forall p \in S, P(p)$. This denotes a mode of specification and proof, not an algorithm that enumerates infinitely many inputs in finite time. Section E.

Injective — A map $f:X\to Y$ is injective if $\forall x1,x2 \in X$, $f(x1) = f(x2)$ implies $x1 = x2$. Section H applies this definition to the carry map K. See [3, §12.2].

**Label** — The numerical identifier p retained in a record. It is distinct from event location $x_p$. Section b.2; paper definition.

**Lean / Mathlib** — Lean is an interactive theorem prover and programming language [5]. Mathlib is a community-developed library of formalized mathematics; its architecture and collaborative development are described in [8]. That 2020 publication predates Lean 4 and is cited for library background, not as evidence of this manuscript's build or proof coverage. The companion README identifies the archived source, library revision, and exact coverage of this paper's formal development.

Local closure — In the prime-record model, incidence and matching put every admitted event at its identified parent midpoint (Theorems 2–3). In the analytic readout, Theorem 8 gives exact recovery of zeros already on the signed crease, with multiplicity. These domains are distinct. Original-zero recovery in Theorem 9 retains the real offset; Section K proves the universal placement equivalence, not universal placement itself.

**Matching / admission rule** — An incident pair is matched when $t = u$. Together, $t+u = 1$ and $t = u$ imply $t = u = ½$. Matching is an admission condition of this model, not a conclusion of the arithmetic definition of a prime. Sections b.3 and D.

**Midpoint / crease** — The location $(a_p+b_p)/2$. "Crease" is this paper's name for the parent midpoint; the term alone asserts no physical fold. Sections b.4 and D; paper convention.

Off-crease — In the prime-record model, an event differs from its identified parent midpoint. In the analytic common frame, an input has $\mathrm{Re}\, s \neq ½$, equivalently real offset $d \neq 0$. Theorem 3 excludes off-midpoint admitted matched records; it does not establish universal critical-line placement of zeta zeros. Sections E, I, and K.

**Parent / parent span** — A parent comprises an identifier $\kappa$ and endpoints $a,b \in \mathbb{R}$ with $a<b$. Its span is $[a,b]$, and its length is $L = b-a>0$. Both readings of an event must use this same identified parent and this same L. Two distinct identifiers may have equal endpoint coordinates without representing the same parent. Sections b.2 and E.

**Participant / projector** — A participant identifier distinguishes an operand or perspective retained in a comparison record. In the paper's one-dimensional representation, the two projector readings are $E(t) = a+(b-a)t$ and $T(u) = b-(b-a)u$, for $t,u \geq 0$. These parameterizations are the mathematical objects used here; the archival perspective terminology does not supply an additional receiver or reflection operation. Sections b.4 and D.

**Premise / theorem / proof** — A premise is an input condition; a theorem is a statement derived from specified conditions; a proof supplies that derivation. Used throughout; standard logical terminology.

**Prime** — For $p \in N_0$, p is prime if $p \geq 2$ and every positive divisor d of p satisfies $d = 1$ or $d = p$. Equivalently, no $u,v \in N_0$ with $u,v \geq 2$ satisfy $p = uv$. Here d divides p when $p = dk$ for some $k \in N_0$. Section C; [1, §1.1.1].

**Projection** — A map retaining one component of a pair, such as $(p,1/2)\mapsto p$. Section E; standard mathematics.

**Ray** — Here a parameterized half-line $E_p(t) = a_p+L_p\,t$ or $T_p(u) = b_p-L_p\,u$ for nonnegative parameters. Parameters are coordinates, not time. Section D; mathematical representation introduced here.

Real offset — At an original zero $\rho = X + iY$, define $d = X - ½$. Its full address is $(Y,d)$, and $C(Y) + d = \rho$. The same-height difference $\rho - C(Y)$ equals d and vanishes exactly when $X = ½$. Section K.

**Record** — A retained record Q contains a numerical label p, participant identifiers, a parent identifier, its endpoints $a<b$, and an event address x. Its normalized readings are derived from these fields. $Q_n$ denotes a record at stage n; $H_n = (Q_0,\ldots,Q_n)$ denotes the ordered history in the worked example. A record and a history are different objects. Equal event addresses do not imply equal records. Sections b.2 and h.1.

**Relative progress / normalized reading** — $t_p = (x_p-a_p)/L_p$ or $u_p = (b_p-x_p)/L_p$. Normalization divides by the same nonzero parent length. These are directed coordinates; inside the segment they are fractions measured from opposite ends. Section b.3; paper definition. See [2, §2.6].

**Retention / inherited state** — An inherited state h is retained by K when a specified decoder R satisfies $R(K(h)) = h$. Additional experience e may be stored with h, but the recovery claim concerns h. Merely keeping a label or an address is insufficient unless it determines the entire state being claimed as recoverable. Section H.

Riemann zeta function and Riemann hypothesis — Initially $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for Re $s > 1$, continued meromorphically with a pole at $s = 1$. The trivial zeros are $-2, -4, -6, \ldots$. The nontrivial zeros lie in the open strip $0 < \mathrm{Re}\ s < 1$. The Riemann hypothesis asserts $\mathrm{Re}\ \rho = \frac{1}{2}$ for every nontrivial zero $\rho$. [4, 7]

In the analytic realization, write $s = X + iY$. The signed chart $C(Y) = \frac{1}{2} + iY$ has one free real parameter. The positive bent path agrees with it for $Y \geq 0$. The readout $F(Y) = \zeta(C(Y))$ covers precisely the crease zeros. The aggregate transform recovers every original strip zero; its full address retains (Y,d). Coverage by Y alone is the full-strip placement condition in Section K.

**Shared parent** — The same identified parent interval is used by the compared records. Distinct parents with equal endpoint coordinates have the same numerical midpoint, but remain distinct parents. Equal normalized readings alone do not establish a common parent or address. Section E; paper condition.

Signed height window — For each $T > 0$, both counts in t.8 use the open window $-T < \mathrm{Im}\ \rho < T$, or $-T < Y < T$ on the crease. Endpoints are excluded, and zeros are counted with multiplicity. Equality at every cutoff is equivalent to universal crease placement. Section p.2.

**Survivor / surviving family** — $S = \mathbb{N}_0 \setminus (\{0,1\} \cup C)$, with C the set of proper products. A survivor is an element $n \in S$. The Prime Survivor Equivalence Theorem establishes $n \in S$ if and only if n is prime. Sections b.1 and C.

**Union / coverage** — The generated collection is $G = \bigcup_{n \in \mathbb{N}_0} G_n$: $q \in G$ exactly when $q \in G_n$ for some finite n. Coverage of primes is the additional statement $\forall p \in S, \exists q \in G$ with label(q) = p. Preservation of validity does not establish this additional statement. Section G. See [3, §1.8].

**VPA** — VPA expands to Value, Perception, Action in the TRYX source vocabulary. It is contextual terminology in this paper, not an additional axiom of Theorems C–H. Their operative objects are explicitly defined records, readings, matching conditions, and recovery maps. No independent mathematical definition of the full native VPA system is established here. Section b.4.

**Whole numbers / nonnegative integers** — 0,1,2,..., denoted $\mathbb{N}_0$. Domain of candidate labels. Section b.1; standard mathematics.

X line / signed Y chart — The prime-record construction uses an affine real X line (Section b.2). Its analytic realization uses complex inputs $s = X + iY$ and the one-parameter signed crease $C(Y) = \frac{1}{2} + iY$ (Section I). A path constraint fixes its own image; full-address recovery retains the real offset required for arbitrary original zeros. Section K.

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**Project sources and verification records**

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[17] Repaired Lean V5. Prime-Crease source, base source, and verification notes.

# Q. Formal verification and source notes

Formal verification. The pinned V5 source enforces nonnegative bent-path distance with a subtype and uses a separate real parameter for the signed chart. Its archived note reports compilation of LocalClosure.lean and PrimeCrease.lean with warnings treated as errors, using Lean 4.34.0-rc2 and Mathlib revision 85e3a25e006c35636f0e53b0e9296caca2685bc0. Reported axioms: propext, Classical.choice, Quot.sound. This paper revision records no fresh compiler run. Theorems 4–6 have written proofs; the formal correspondence table identifies the other covered statements and preserves their hypotheses. [17]

| Claim | Lean declaration |
|---|---|
| Common-frame crease under its stated hypothesis | common_crease |
| Turned path evaluation | bent_zeta_readout |
| Exact crease zeros and multiplicity | readout_exact, readout_preserves_analytic_order |
| Prime spectrum and weighted series | prime_spectrum_agrees, aggregate_agrees |
| Simple poles and residue form | aggregate_simple_pole_iff, aggregate_residue_form |
| Full address and placement equivalence | decode_encode, full_coverage_iff_placement |

Reproduction. Retrieve both Lean files and the verification note at the pinned links in reference [17]. In the recorded Mathlib environment, compile LocalClosure.lean first, then PrimeCrease.lean with the local directory in LEAN_PATH, following that note’s exact command. The newly assembled publication project still needs a clean build. For the bounded contour application, reference [16] supplies the retrieved report; the exact rational certificate and executable verifier remain pending retrieval. No fresh contour replay is claimed.